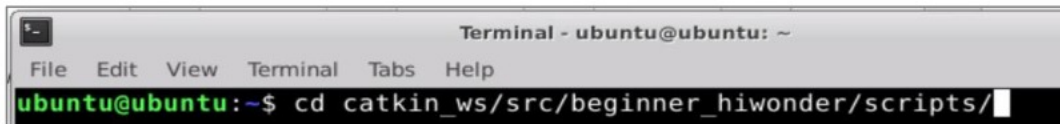


第 8 课 编写简单的客户端 Client

本节以创建简单的服务（Client）节点turtle_spawn.py为例进行讲解。该节点通过程序的形式发布一个让客户端产生一个新的小乌龟的请求。

1.配置客户端代码编译规则

- 1) 输入指令“`cd catkin_ws/src/beginner_hiwonder/scripts/`”，回车。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ cd catkin_ws/src/beginner_hiwonder/scripts/
```

- 2) 输入指令“`vi turtle_spawn.py`”编辑程序，复制下面程序。如需修改，再按下“i”即可修改。修改完成，按下“Esc”，输入“: wq”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ vi turtle_spawn.py
```

```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

# 该例程将请求/spawn服务，服务数据类型turtlesim::Spawn

import sys

import rospy

from turtlesim.srv import Spawn
```

```
def turtle_spawn():

    # ROS节点初始化

    rospy.init_node('turtle_spawn')

    # 发现/spawn服务后，创建一个服务客户端，连接名为/spawn的service

    rospy.wait_for_service('/spawn')

    try:

        add_turtle = rospy.ServiceProxy('/spawn', Spawn)

        # 请求服务调用，输入请求数据

        response = add_turtle(2.0, 2.0, 0.0, "turtle2")

        return response.name

    except rospy.ServiceException, e:

        print "Service call failed: %s"%e

if __name__ == "__main__":

    #服务调用并显示调用结果

    print "Spwan turtle successfully [name:%s]" %(turtle_spawn())
```

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
6
7 import sys
8 import rospy
9 from turtlesim.srv import Spawn
10
11 def turtle_spawn():
12     # ROS节点初始化
13     rospy.init_node('turtle_spawn')
14
15     # 发现 /spawn服务后，创建一个服务客户端，连接名为 /spawn的 service
16     rospy.wait_for_service('/spawn')
17     try:
18         add_turtle = rospy.ServiceProxy('/spawn', Spawn)
19
20         # 请求服务调用，输入请求数据
21         response = add_turtle(2.0, 2.0, 0.0, "turtle2")
22         return response.name
23     except rospy.ServiceException, e:
24         print "Service call failed: %s"%e
25
26 if __name__ == "__main__":
27     #服务调用并显示调用结果
28     print "Spawn turtle successfully [name:%s]" %(turtle_spawn())
:wq
```

3) 输入指令“`chmod +x turtle_spawn.py`”回车，为保存的turtle_spawn.py赋予可执行权限。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x turtle_spawn.py
```

2.运行客户端

1) 输入指令“`roscore`”，启动节点管理器。

```
ubuntu@ubuntu:~/catkin_ws$ roscore
```

若已开启，则会出现以下提示：

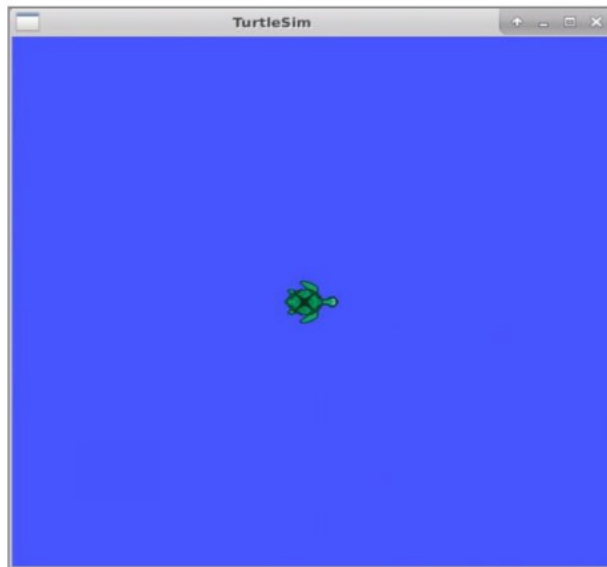
```
RLException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://ubuntu:11311/
The traceback for the exception was written to the log file
```

2) 输入指令“`roslaunch turtlesim turtlesim_node`”，回车，开启小乌龟。

```
ubuntu@ubuntu:~/catkin_ws$ roslaunch turtlesim turtlesim_node
```



此时，界面会弹出小乌龟运行窗口，如下图所示：



3) 重新打开一个命令行终端,输入指令“**roslaunch beginner_hiwonder turtle_spawn.py**”
回车，运行客户端。

```
ubuntu@ubuntu:~$ roslaunch beginner_hiwonder turtle_spawn.py
Spwan turtle successfully [name:turtle2]
ubuntu@ubuntu:~$
```

这时，客户端发送请求至服务端，并应答开启另一个小乌龟，如下图所示：

